



**Bureau of Experts at the Council of Ministers
Official Translation Department**

Unified Law of Healthcare Waste Management in the GCC States

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Translation of Saudi Laws

NOTES:

1. This translation is provided for guidance. The governing text is the Arabic text.
2. The translation of Saudi laws takes the following into consideration:
 - Words used in the singular form include the plural and vice versa.
 - Words used in the masculine form include the feminine.
 - Words used in the present tense include the present as well as the future.
 - The word “person” or “persons” and their related pronouns (he, his, him, they, their, them, and who) refer to a natural and legal person.

For any comments or inquiries, please contact the Official Translation Department at:

otd@boe.gov.sa



Unified Law of Healthcare Waste Management in the GCC States

Chapter 1: General Provisions

Article 1: Definitions

In the application of the provisions of this Law, the following words and phrases shall have the meanings assigned thereto, unless the context requires otherwise:

1. **Healthcare Waste:**

Waste generated from facilities that provide healthcare services; laboratories; centers for producing medicines, pharmaceuticals, and vaccines; veterinary treatment centers; research institutions; and home-based treatment and nursing.

Such waste shall be divided into two types:

1.1 Non-hazardous healthcare waste:

All waste that includes materials similar to those found in municipal waste. Such waste is generated from administrative departments and from general cleaning work within healthcare facilities, and constitutes the largest portion of total healthcare waste. This type of waste shall be treated as municipal waste.

1.2 Hazardous healthcare waste:

Waste generated from sources that are contaminated or potentially contaminated with infectious, chemical, or radioactive agents. Such waste constitutes the lowest percentage of total healthcare waste and poses a risk to individuals, the community, and the environment during its production, collection, handling, storage, transport, or disposal.

2. **Pathological Waste:**

Waste containing human tissues, organs, embryonic and placental tissues, animal carcasses, blood and its derivatives, and bodily fluids.

3. **Infectious Waste:**

Waste that may transmit any infectious diseases as a result of its contamination with bacteria, viruses, parasites, and fungi.

4. **Chemical Waste:**

Waste that contains solid, liquid, or gaseous chemical substances resulting from diagnostic, therapeutic, or laboratory activities, or used in cleaning, disinfection, or sterilization processes.

5. **Sharps Waste:**

Waste that contains sharp tools, such as syringes, scalpels, surgical scalpels, saws, blades, shattered glass, and any other sharp tools which may cause cuts or punctures to the body.

6. **Pharmaceutical Waste (Medicines):**

Waste generated from preventive or therapeutic activities, from the production of pharmaceutical preparations, and from damaged or expired pharmaceuticals and medicines.



7. Waste Contaminated with Radioactive Materials:

Waste that contains all solid, liquid, and gaseous materials contaminated with radionuclides resulting from their use in examinations of human tissues and fluids, and in diagnosing, identifying, and treating tumors.

8. Compressed Gas Containers Waste:

Empty or damaged gas cylinders, cartridges, sterilization containers, and aerosols.

9. Genotoxic and Cytotoxic Waste:

Waste of specific pharmaceutical substances that are extremely hazardous and have the ability to kill or prevent the division of cells or damage genetic materials. Such waste contains substances used in the treatment of certain types of cancer and in organ transplant as well as any supplies used in the preparation of such substances, in addition to secretions of patients treated therewith, for up to a week from the date of the patient's last administered dose.

10. Waste Treatment Unit:

The facility where the processes of altering the biological, chemical, or physical properties of hazardous healthcare waste are carried out with the aim of eliminating its hazard so that it becomes safe for health and the environment.

11. Waste Segregation (Separation):

Separating waste into different categories (by the generator) starting from the point of generation in the facility and during the stages of collection, packaging, storing, and transport within the facility.

12. Storage:

Temporary holding of hazardous healthcare waste at a specific collection site.

13. Transporter:

A natural person or a legal person (a company or public or private establishment) that engages in the transportation of hazardous healthcare waste to the treatment and disposal unit.

14. Transport Document:

The form that includes all the information and is signed by the generator, transporter, and disposer; it often consists of carbon copies which shall accompany each load of healthcare waste that is transported from the waste generating facility to the treatment unit.

15. Facility:

Any hospital, clinic, medical or veterinary center, pharmaceutical company, medical research center, pharmacy, laboratory, or public or private convalescent home.

16. Generator:

Any natural person or legal person, such as a health facility, whose activity results in the generation of healthcare waste.

17. Competent Authority:

The environmental authority or any other entity designated by the State.

18. Waste Incineration:

The process of disposing of combustible solid, liquid, and gaseous wastes at high temperatures, which produces gases, substances, or compounds that do not affect the environment, and residues that do not contain hazardous substances.



Article 2: Objective and Scope of the Law

This Law aims to establish an appropriate method for controlling and monitoring the generation, segregation, storage, transport, treatment, and disposal of hazardous healthcare waste in a safe manner in the GCC States.

It also aims to support and develop the processes of treatment of such waste in a manner that protects public health and prevents environmental pollution.

This Law shall apply to every generator, transporter, or disposer engaged in the collection, storage, transport, treatment, or disposal of hazardous healthcare waste. Radioactive materials shall be handled in accordance with the Unified Law for Handling Radioactive Materials in the GCC States.

Chapter 2: Requirements for Generators of Hazardous Healthcare Waste

Article 3

Generators of hazardous healthcare waste must reduce the generation rates of such waste, in terms of quantity and quality, by using up-to-date devices and equipment, adopting clean technology, and choosing alternatives and primary materials that are less harmful to the environment and public health. Each health facility must develop a comprehensive work program for proper waste management.

Article 4: Classification

Hazardous healthcare waste shall be classified as follows:

1. Infectious Waste:

Waste containing pathogens, including bacteria, viruses, parasites, or fungi, in quantities or concentrations sufficient to cause diseases to individuals exposed to such pathogens. Such waste includes the residues of microbial cultures, surgical waste, waste from isolation departments, and waste from dialysis departments of patients with infectious diseases.

2. Pathological Waste:

Waste containing tissues, organs or parts thereof, embryonic and placental tissues, blood and its derivatives, other bodily fluids, and animal carcasses.

3. Sharps Waste:

Waste that may cause cuts, wounds, or punctures to the body, such as syringe needles, scalpels, blades, broken glass, and other sharp instruments.

4. Pharmaceutical Waste (Medicines):

Waste resulting from the manufacture and preparation of medicines and pharmaceutical preparations as well as from damaged or expired pharmaceutical products, contaminated pharmaceutical products, and sera and vaccines, including containers and tools used in the production, packaging, and distribution thereof.



5. Genotoxic and Cytotoxic Waste:

Waste that can affect genes and cells, leading to health problems such as congenital abnormalities in fetuses, or that is capable of causing cancer and inhibiting cell growth. Substances producing such waste are used in nuclear medicine departments, oncology units, and diagnostic radiology units, and are present in sewage from toilets used by patients treated therewith.

6. Chemical Waste:

Chemical waste shall be deemed hazardous if it is:

- toxic;
- ignitable;
- corrosive to other substances;
- reactive or explosive; or
- capable of causing congenital abnormalities in fetuses, altering genetic material, causing cancer, or inhibiting cell growth.

7. Radioactive Waste:

All radioactive materials, whether solid or liquid, used in examination, diagnosis, and treatment, as well as all solid or liquid materials contaminated therewith.

8. Compressed Gas Containers Waste:

Empty or damaged compressed gas containers that are used for the containment of inert gases or gases that are likely to cause harm. Such containers may explode if punctured or exposed to high temperatures.

Article 5: Requirements for the Segregation (Separation) and Packaging of Hazardous Healthcare Waste

- The generator of hazardous healthcare waste shall separate such waste from non-hazardous healthcare waste at its points of generation. The generator shall bear direct responsibility for segregation (separation) and packaging at the sites designated for such purpose within the health facility and medical departments as follows:

1. Infectious medical waste shall be collected in yellow plastic bags bearing the words "HAZARDOUS MEDICAL WASTE" and the biohazard symbol; such bags shall conform to the specifications provided for in Annex 4.
2. Sharps waste shall be collected in thick, yellow, puncture-resistant, and leak-proof containers bearing the words "SHARPS WASTE" and the biohazard symbol.
3. Liquid chemical waste shall be collected in thick, yellow, tightly sealed leak-proof containers bearing the words "CHEMICAL WASTE". Solid chemical waste shall be collected in yellow plastic bags bearing the words "CHEMICAL WASTE - MEDICINES" and the biohazard symbol.
4. Pharmaceutical waste (medicines):
 - 4.1 Expired medicines and pharmaceutical materials, if found in large quantities, shall be returned to the pharmacy department for disposal through proper methods.
 - 4.2 Residues of medicines and pharmaceutical materials that might be contaminated shall be disposed of by placing them first in leak-proof containers, and then in yellow plastic bags



bearing the medicines symbol and the biohazard symbol.

5. Radioactive waste shall be collected in containers manufactured for such purpose, according to the specifications set by the competent authority, made of or encased with lead, and tightly sealed. Such containers shall bear the international radiation symbol.
6. Parts and remains of human organs shall be collected in red plastic bags bearing the biohazard symbol and kept in the mortuary refrigerator until handled in accordance with fatwa or as determined by each State.
7. Highly infectious hazardous waste produced from microbial cultures shall be collected in autoclavable plastic bags within the generating departments. Such bags shall, after primary treatment using an autoclave, be placed in yellow bags bearing the words "HAZARDOUS MEDICAL WASTE" and the biohazard symbol.
8. Genotoxic and Cytotoxic waste shall be collected in yellow leak-proof containers bearing the words "RESIDUES OF CYTOTOXIC SUBSTANCES".

Such waste shall be returned to its source or incinerated at very high temperatures, 1200 degrees Celsius and above, and shall not be buried, discharged into the sewage network, or mixed with other pharmaceutical materials.

Article 6: Labeling Requirements

1. The generator of hazardous healthcare waste shall, prior to transporting such waste to the storage site within the health facility or the treatment unit, affix adhesive labels to waste containers and bags that contain the following information or print such information thereon:
 - 1.1 Name of the waste generator (name of the facility).
 - 1.2 Name of the site (department or ward).
 - 1.3 Type of generated waste according to the classification provided for in Article 4.
 - 1.4 Weight and quantity of waste contained in the container or bag.
 - 1.5 Time and date of collection.
 - 1.6 Time and date of transport.
2. Labels affixed to containers and bags shall be of an appropriate size, printed in permanent ink, and be water-resistant.
3. The biohazard symbol shall be placed on containers or bags, in accordance with Annex 5.



Article 7: Requirements for On-Site Collection and Transport

1. Hazardous healthcare waste bags and containers shall be collected and transported by trained workers using trolleys designated for such purpose to ensure maximum safety levels during collection and transport operations within health facilities, in order to prevent the contents of such bags and containers from being scattered or leaked.
2. Hazardous healthcare waste bags and containers must, prior to collection and transport, be tightly closed and must bear labels identifying the waste contained therein as well as the biohazard symbol.
3. Waste must not exceed three-quarters of the bag's capacity, be pressed or compacted, or be held against the body or grasped from the bottom when carried. Such bags must be held by the upper part when being carried.
4. On-site transport of hazardous healthcare waste must be done using covered trolleys designated for such purpose and designed to ensure efficiency when loading and unloading. Such trolleys must be robust, leak-proof, and easy to clean and disinfect with disinfectants.
5. Hazardous healthcare waste generated by infectious disease departments and rooms, as well as isolation departments and rooms, shall be collected under the direct supervision of the healthcare waste management officer in the health facility.
6. Human, embryonic, and placental tissues and organs shall be collected separately and kept in the mortuary refrigerator or in a designated refrigerator until disposed of pursuant to the laws applicable in each state in accordance with the principles of Shariah.
7. Animal carcasses and tissues shall be collected and kept separately in a refrigerator until they are treated and disposed of.
8. Trolleys designated for the collection and transport of hazardous healthcare waste shall be cleaned, washed, and disinfected daily at a designated site by trained workers and under the supervision of the healthcare waste officer at the health facility. The residues from such cleaning shall be treated before their discharge or disposal.
9. If hazardous healthcare waste leaks or scatters from bags, containers, or transport trolleys, such waste must be deemed highly hazardous, and the situation must be handled immediately, with disinfection and safety measures implemented at the site of the release.
10. Non-hazardous healthcare waste shall be collected in black bags and handled separately from hazardous healthcare waste at all stages (packaging, collection, on-site transport, and storage) until such waste is transported for final disposal at landfill sites designated by the municipality.



Article 8: Requirements for Temporary On-Site Storage

Each health facility seeking to temporarily store hazardous healthcare waste within the facility until such waste is transported to the treatment unit, shall satisfy the following requirements:

1. A storage site shall be designated within the health facility to serve as a center for the collection of hazardous healthcare waste generated by the facility.
2. Waste shall be packaged inside containers or bags prior to storage.
3. The storage site shall be suitable so as not to cause any pollution or harm to human health or the environment.
4. The storage site shall be located in a building that is securely locked and prevents the leakage of water and seepage of rainwater, the spread of foul odors, and the entry of rodents, insects, birds, or stray animals, with a floor that is solid, durable, capable of withstanding washing and disinfection, and equipped with proper drainage.
5. The storage site shall be equipped with safety and firefighting equipment.
6. The storage site shall be managed by personnel specialized in hazardous healthcare waste management.
7. The storage site shall be well-lit, ventilated, and air-conditioned to maintain a temperature between 15 and 18 degrees Celsius.
8. The storage period for hazardous healthcare waste shall not exceed 24 hours.
9. The storage site shall be easily accessible for the purposes of storage, transport, and cleaning.
10. The storage site shall be located far from kitchens, food preparation and food storage areas, and patient care areas.
11. Access to the storage site shall be restricted to authorized personnel only.
12. Clear signs shall be placed on the storage site indicating its contents.
13. The storage site shall be equipped with appropriate hygiene tools as well as disinfection and sterilization materials to be used continuously for the site's sanitation, and for emergencies and in the event of waste spills.
14. A contingency plan shall be put in place for dealing with waste spills by the personnel responsible for the storage site.

Article 9: Requirements for Generators of Hazardous Healthcare Waste for Off-Site Transport

The generator of hazardous healthcare waste shall comply with the following procedures prior to shipping the waste outside the facility:

1. Proper packaging and labeling of hazardous healthcare waste in accordance with Articles 5 and 6.
2. Ensuring that no shipment of hazardous healthcare waste is handed over to a person or a facility not licensed to transport hazardous healthcare waste by the competent authority.
3. Ensuring that no shipment of hazardous healthcare waste is handed over for off-site transport unless it is accompanied by the transport document (Annex 7).
4. Ensuring that no shipment of hazardous healthcare waste is handed over to a treatment facility that does not hold a permit from the competent authority.



Article 10: Requirements for Generator Reports and Record-Keeping

1. The generator of hazardous healthcare waste shall provide reports on all aspects relating to hazardous healthcare waste, such as information on generation, storage, transport, and treatment.
2. The competent authority shall be provided with a copy of such reports on a periodic basis, as determined by such authority.

Chapter 3: Requirements for Off-Site Transport of Hazardous Healthcare Waste

Article 11: License for Off-Site Transport

Any person or facility transporting hazardous healthcare waste off-site shall obtain a license from the competent authority.

Article 12: Information Required for a Transport License

To obtain a license from the competent authority, the applicant shall provide the following information:

1. A description of the means of transport and the equipment to be used in transport.
2. The contingency plan to be used in case of accidents or waste leakage at handover points or during transport.
3. A list of the names of the workers and their expertise, and a certificate proving they are medically fit to practice such profession, provided that not more than one year has passed since the issuance of the certificate.
4. A training program for workers in the field and an awareness program.
5. Any additional information that the competent authority deems necessary for the protection of human health and the environment.



Article 13: Off-Site Transport

Transporters of hazardous healthcare waste shall comply with the following requirements:

1. Refrain from transporting any waste to a treatment unit that does not have a permit for the disposal of hazardous healthcare waste from the competent authority.
2. Refrain from transporting any chemical waste that is not accompanied by its safety data sheet. (Annex No. 8).
3. Refrain from transporting any waste that is not accompanied by a transport document duly completed by the generator.
4. Refrain from commingling wastes of different shipping specifications by placing them in a single container.
5. Refrain from accepting any container or bag that does not have a label showing the information provided for in Article 6.
6. Provide the competent authority, prior to commencing transport, with the waste transportation schedule, indicating the generator's name, the type and quantity of waste to be transported, and the start and end dates of the transport.
7. Refrain from transporting any container or bag that does not meet the specifications provided for in Article 5.
8. Refrain from storing hazardous healthcare waste without obtaining a permit from the competent authority for storing such waste.
9. Regularly maintain means of transport and equipment to minimize risks to human health and the environment.
10. Refrain from passing through residential areas or commercial streets when transporting hazardous healthcare waste during peak periods.
11. Affix signs to the means of transport indicating the type of materials transported. The transporter shall be fully aware of the degree of hazard of such materials and the measures to be taken in the event of an emergency during transport.
12. Accurately complete the section allocated to them in the transport document.
13. Adhere to the transport times set by the relevant authorities.
14. Maintain records and documents relating to the transport of waste and submit the same to the competent authority upon request within a maximum period of one week from the date of the request.
15. Use means of transport that meet the requirements for vehicles provided for in Annex 3.



Chapter 4: Requirements for Hazardous Healthcare Waste Treatment Facilities

Article 14: Licensing Treatment Facilities

Any person or facility seeking to establish a treatment facility for hazardous healthcare waste shall obtain a license from the competent authority and commit, when choosing a site, to preserving the environment as well as the integrity of groundwater and surface water and the quality of the ambient air, and not causing harm to any of the foregoing or to neighboring residential areas. The applicant shall, to obtain a license from the competent authority, complete the environmental impact assessment data form approved by each State.

Article 15: Information Required for Licensing Treatment Facilities

To obtain a license for a treatment facility, the applicant shall provide the following information:

1. A detailed description of the technologies and methods to be used in the treatment of hazardous healthcare waste, including technology specifications, waste treatment methods, and emission control measures, as well as the quantity of the materials generated after treatment and the methods of disposal thereof.
2. The types of hazardous healthcare waste to be treated by the technology or technologies for which licensing is sought.
3. A detailed description of the design, construction, and operation of the facilities.
4. A map showing the geographical and geological location of the facilities.
5. A contingency plan to be used in cases of waste leakage during treatment or disposal, including a list of emergency equipment.
6. Procedures for self-inspection and maintenance of facilities and equipment used in waste treatment.
7. Any other information requested by the competent authority for the issuance of an environmental license.



Article 16: Establishment and Operation of Treatment Facilities

Any person or facility seeking to establish and operate a unit for the treatment of hazardous healthcare waste shall comply with the following:

1. Obtain approval for the treatment technology from the competent authority in the State. An indicative list of currently available technologies is provided in Annex 2.
2. Apply environmental protection standards for air, water, and waste, with reference to Annex 1.
3. Treat any liquid materials that may result from treatment.
4. Ensure that the performance efficiency and removal rate of the technology used are not less than 99.99%.
5. Develop and implement a training program for facility workers in the field of hazardous healthcare waste management.
6. Submit certificates proving that the workers are medically fit, subject to annual renewal.
7. Provide certificates of experience for the workers, if any.
8. Refrain from accepting any hazardous healthcare waste that is not accompanied by a transport document completed by the generator and the transporter.
9. Refrain from accepting any hazardous healthcare waste from a transporter who does not have a transport permit from the competent authority.
10. Refrain from accepting any waste that is not accompanied by the information specified in the labeling requirements provided for in Article 6.
11. Ensure that each shipment of waste received at the facility conforms to the specifications stated in the transport document accompanying such shipment.
12. Notify the competent authority of any change in the ownership and operational management of the facilities.
13. Dispose of the residues and waste resulting from treatment operations at the disposal sites designated by the competent authority.
14. Refrain from accepting hazardous healthcare waste that cannot be treated at the facility using available treatment technologies according to the license granted thereto.



Article 17: Operational Record

When operating a treatment facility for hazardous healthcare waste, the applicant shall commit to the following:

1. Maintain an operational record that includes the following:
 - 1.1 A description of the type and quantity of each shipment received, the name of the generator recorded in the transport document, the date of receipt, and the date of treatment.
 - 1.2 The type and results of analyses of the waste resulting from treatment.
 - 1.3 The type and results of the efficiency tests of the treatment device.
 - 1.4 Copies of the waste transport documents.
 - 1.5 Copies of all safety data forms for each waste shipment.
 - 1.6 Measurements of emission concentrations in the air resulting from treatment.
 - 1.7 The amount of waste resulting from treatment, and the method and location of disposal thereof.
 - 1.8 Results of the analysis of wastewater resulting from treatment, and the methods and location of disposal thereof.
 - 1.9 Any other records the competent authority may deem necessary to retain.
2. Submit quarterly reports on the entire operational record to the competent authority.

Article 18: Monthly Report

The operator of the treatment facility shall, upon the request of the competent authority, submit a monthly report containing the daily quantities of waste received from each generator separately, and indicating the name of the generator and the transporter.

Chapter 5: Cross-Border Transport

Article 19

Cross-border transport of hazardous healthcare waste shall be carried out in accordance with regional and international agreements, the coordination procedures among the GCC States regarding waste transport across their borders, as approved by the leaders of the GCC States, and the relevant national laws.



Chapter 6: Penalties

Article 20: Issuance of State-Specific Laws

Each State shall enact its own law to penalize any person who violates the provisions of this Law.

Chapter 7: Final Provisions

Article 21: Scope of Requirements

The competent authorities shall determine the scope of the requirements stipulated in this Law, as well as any new requirements that may arise and any amendments or updates thereto.

Article 22: National Legislations and Laws

This Law shall constitute the minimum standard for national legislations and laws relating to the sound management of hazardous healthcare waste.

Article 23: Annexes

The attached annexes shall be deemed an integral part of this Law.



Annexes

Annex 1

Guidelines governing standards for emissions from the incineration of hazardous healthcare waste

Contaminant	Measurement
Total Suspended Particulates (TSP)	34 mg/m ³ (1) (adjusted to 7% oxygen)
Opacity	10% with the exception of a 6-minute period during any hour
Carbon Monoxide (CO)	50 mg/m ³
Sulfur Dioxide (SO ₂)	150 mg/m ³
Hydrogen Chloride (HCl)	100 mg/m ³ or at least 97% removal
Nitrogen Oxides (NO _x)	400 mg/m ³
Organic Compounds	8 ppm or at least 99.99% removal
Hydrogen fluoride (HF)	5 mg/m ³
Dioxins and Furans	125 ng/m ³
Cadmium (Cd)	0.16 mg/m ³
Lead (Pb)	1.2 mg/m ³
Arsenic (As)	1.2 mg/m ³
Mercury (Hg)	0.55 mg/m ³

- Standard dry cubic meter.
- The stack height must be 2.5 times the height of the nearest neighboring building.
- Combustion efficiency must be demonstrated by calculation prior to any waste incineration process, taking into account the heating (calorific) value and the physical and chemical properties of the waste to be incinerated.
- These standards shall apply to healthcare waste incinerators with a design capacity of 500 kg/hour or more.



Annex 2

Common technologies for treating hazardous healthcare waste

Treatment Technology	Waste That Can Be Treated	Advantages	Disadvantages
Chemical Treatment	-Infectious waste - sharps	- An effective method to reduce the pathogenic organisms (pathogens) in hazardous medical waste.	- Used disinfectants are hazardous chemicals that must be stored and handled with care. - Not suitable for treating other types of hazardous healthcare waste. - Does not reduce waste volume unless the waste is shredded beforehand.
Microwave	Infectious waste - sharps	- Gases emitted into the external environment are considered very low. - Reduces waste volume by about 75% through shredding.	- Requires shredding waste and moistening it with water. - Not suitable for treating other types of hazardous healthcare waste.
Autoclave	- Infectious waste - sharps	- Effective in reducing pathogenic organisms (pathogens) in hazardous medical waste. - Low operational costs.	- Not suitable for treating other types of hazardous healthcare waste. - Does not reduce waste volume unless the waste is shredded beforehand.
Incineration	- All hazardous healthcare waste except radioactive waste, compressed gas containers, and liquid waste.	- Suitable for treating most types of hazardous healthcare waste. - Greatly reduces waste volume.	- Gases emitted as a result of combustion are very high, which necessitates the use of advanced, high-cost technologies to control gas emissions.



Annex 3

Vehicle Requirements

The transporter shall use hazardous healthcare waste transport vehicles that meet the following requirements:

1. The vehicle shall have a liquid-tight waste compartment designated for transporting hazardous healthcare waste.
2. Compactor vehicles shall not be used in the transport of hazardous healthcare waste.
3. The vehicle shall be equipped with the following:
 - a) Disinfectants suitable for use in the event of leakage.
 - b) Fire extinguishers readily accessible and located in a separate place from the waste compartment, in a number commensurate with the size of the vehicle and the space available therein.
 - c) The surfaces of transport vehicles shall be manufactured of materials that are rust-resistant, non-absorbent, and corrosion-resistant, and that are washable and capable of being disinfected with appropriate chemicals.
4. The waste compartment of the vehicle shall be disinfected after each transfer and in the event of any leakage, at a location equipped for such purpose at the treatment site, in accordance with the instructions set by the Ministry of Health in coordination with the competent authority.
5. The waste compartment shall be securely locked.
6. Hazardous healthcare waste shall be completely isolated from the driver's cab.
7. The use of ordinary open-bed vehicles is prohibited.
8. Vehicles shall bear clear markings and information on the sides and at the rear of the cargo compartment indicating the hazardousness of the cargo. Such markings shall be in large and clear letters not less than 3 inches in height.
9. Transport vehicles shall be fitted with all required safety equipment and be in good working condition.
10. Trailers may be used to transport waste provided that they meet the above requirements.
11. Vehicles shall not be used for storage purposes.



Annex 4

Specifications of Plastic Bags and Containers

First:

Bags and sharps containers must not contain halogenated plastics or PVC.

Second:

Specifications of plastic bags:

1. Plastic bags used to collect hazardous healthcare waste shall be available with the following specifications:
 - 1.1 Plastic bags shall have a thickness of not less than 150 microns and shall have ties for sealing the bag.
 - 1.2 The maximum total capacity shall be 100 liters.
 - 1.3 Bags dimensions shall be compatible with the dimensions of the container in which it is placed during use.
 - 1.4 Bags color shall match the recommended color in the attached table, which shows the recommended distinctive colors for medical waste bags and containers.
2. Bags used to collect highly infectious waste and require initial treatment with an autoclave, for example, must be made of a plastic material that withstands high temperatures without melting, and it is recommended to use plastic made of polyethylene-polyamide (PE/PA).

Third:

Specifications of containers in which bags are placed during use:

1. Containers shall be of a size that accommodates the bag used.
2. Containers shall have a tight-fitting, foot-operated lid.
3. Containers shall be easy to clean and made of disinfectable material.
4. Containers shall be equipped with handles for ease of transport.
5. Containers used for yellow bags shall bear the words “HAZARDOUS MEDICAL WASTE” on their sides and on their lid.
6. Containers shall be easy to move and equipped with wheels.



Fourth:

Specifications of sharps waste containers:

Sharps waste shall be collected in containers designated for such purpose. Specifications of the sharps waste disposal container shall be as follows:

1. Containers shall be made of puncture-resistant material that is impermeable to liquids.
2. Containers shall be fitted with a tight-fitting lid with an opening that allows the insertion of sharp tools, such as syringes, scalpels, etc.
3. Containers shall be yellow in color and shall bear the biohazard symbol and the words “DANGER - SHARPS WASTE”.
4. Containers shall be of a size that allows one-handed carrying and be equipped with a handle for this purpose.

Fifth:

Specifications of radioactive waste containers:

1. Special containers made of lead and encased with lead, in accordance with the specifications determined by the competent authority in each State, and bearing the international radiation symbol.
2. Chemical liquids used for developing radiographic films shall be collected in securely sealed containers for treatment and subsequent beneficial use, subject to the approval of the competent authority.



Sixth:

Specifications of trolleys for on-site transport of healthcare waste:

1. Trolleys shall be manufactured of a rust-resistant material that is not affected by acids and alkalis.
2. Trolleys shall be leak-proof.
3. Trolleys shall be equipped with a lid that can be securely sealed.
4. Trolleys shall have surfaces and corners that are easy to clean.
5. Trolleys shall be of a size sufficient to carry a maximum of ten bags at a time to facilitate movement within hospital departments.
6. Trolleys shall be easy to maneuver and equipped with wheels.

Recommended Distinctive Colors for Healthcare Waste Bags and Containers

Type of Waste	Color of Bag or Container
Highly Infectious Waste	Yellow bearing the words “HIGHLY INFECTIOUS WASTE” and the biohazard symbol.
Human Parts and Organs Waste	Red bearing the words “HAZARDOUS WASTE” and the biohazard symbol.
Infectious Waste	Yellow bearing the words “HAZARDOUS WASTE” and the biohazard symbol
Sharps Waste	Yellow bearing the words “SHARPS WASTE ONLY” and the biohazard symbol.
Chemical and Pharmaceutical Waste	Yellow bearing the biohazard symbol
Radioactive Waste	Securely sealed container bearing the international radiation symbol
Non-Hazardous Medical Waste	Black



Annex 5

International Symbols



International Nuclear Radiation Symbol



Biohazard Symbol



Annex 6

Safety and Health Measures

Owners, operators, officials of health facilities, transport companies, and treatment and disposal sites for healthcare waste shall follow safety procedures in handling, storage, treatment, and disposal, and shall comply with the following requirements:

1. Obtaining complete and up-to-date information about the properties of the waste they handle.
2. Implementing training and awareness programs for workers in such facilities.
3. Training workers to perform their duties with a high level of efficiency.
4. Raising workers' awareness by providing the necessary and important information about the hazards of the materials they handle, and by informing them of the methods of prevention, and raising the awareness of patients and the public when the situation so requires.
5. Establishing and developing occupational health programs for workers that contain the following:
 - 5.1 Initial medical examinations.
 - 5.1.1 Periodic medical examinations conducted annually or as necessary.
 - 5.1.2 Special medical examinations according to the needs of workers at the time.
 - 5.2 Notifying the health authorities when any disease or infection resulting from exposure to healthcare waste is detected.
 - 5.3 Informing the health authorities of occupational injuries, especially punctures or cuts with sharp objects, using the form prepared by the competent authority.
 - 5.4 Vaccination against diseases to which workers are exposed, such as:

Hepatitis B, tetanus, pulmonary tuberculosis, diphtheria, typhoid, or any other diseases as determined by the competent authority.
 - 5.5 Ensuring workplace safety procedures.
 - 5.5.1 Providing changing rooms and bathrooms to ensure personal hygiene.
 - 5.5.2 Providing personal protective equipment and obliging workers to use it.
 - 5.5.3 Placing symbols that warn of the type and hazardousness of the material, such as the nuclear radiation symbol, the biohazard symbol, or the sharps symbol.
6. Training workers to prepare for and respond to emergencies in the event of accidents that expose workers and the public to the hazards of healthcare waste.
7. Preventing leakage or scattering of waste:

Care must always be taken to prevent the leakage and scattering of waste. If waste leaks or scatters, the contaminated area must be cleaned and disinfected immediately. The competent authority must also be notified of this matter.



8. Precautionary measures in the event of emergency:

Infection resulting from accidents during the performance of daily work tasks can be avoided by training workers at sites outside health facilities, such as storage locations and treatment centers, as well as during transport, to conduct first aid and observe precautionary measures to deal with the consequences of accidents associated with these activities. The following are examples of accidents resulting from cuts, burns, fire, and smoke that require first aid, and the treatment measures taken in respect thereof:

- 8.1 Incised skin wounds: In the event a worker sustains an incised skin wound, the case must be referred to the treating specialist immediately after washing the wound with running water and pressing it to expel blood.
- 8.2 Burns: Burns shall be washed with a large amount of running water for more than 15 minutes, then the patient shall be referred to the specialist doctor.
- 8.3 Fire, Smoke, etc.: In case of emergency due to ignition, fire, or smoke from waste, the fire shall be extinguished as quickly as possible to prevent the spread of damage. Full training in the use of fire extinguishers is recommended.



Annex 7

Hazardous Healthcare Waste Transport Document Data Form

(Please print or write all data in a clear font)

Country Name:

Serial No.:

Competent Authority:

Health Facility:

Reference:

A- (1) The waste mentioned in (a)(2) below is collected from

To transport to

Name:..... Signature.....

Waste Position: On behalf of:

Source Telephone No.: Address:.....

Certificate Waste Collection Date:

A- (2) Description of Waste (Quantity and Category)

.....
.....

I hereby certify that I have received the waste shipment and that the information mentioned in A- (1) and A- (2) is correct, subject to any modifications.

I list them as follows:

Waste I received this shipment on Time:

Transporter Name:..... Signature:..... Date:.....

Certificate Truck license plate No. Telephone No.:

Address:.....

Name and address of the institution:

This waste was delivered by a truck with license plate No. at exactly o'clock

on The transporter stated that his name is

Treatment acting on behalf of I hereby certify that the information mentioned in (a)(2) and amended if necessary in (b) above is correct, subject to any modifications. I list them as follows:

Facility Name:..... Signature..... Position:.....

Certificate Date:..... On behalf of



Annex 8

Material Safety Data Sheet (MSDS)

Chemical Name:

Trade Name:

Hazard Class (UN No.):

Serial No. * (CAS No.):

Raw Materials:

Uses:

Physical and Chemical Properties:

Exposure Hazards:

Fire Extinguishing Measures:



First Aid:

Packaging, Storage, and Treatment Methods:

Notes:
